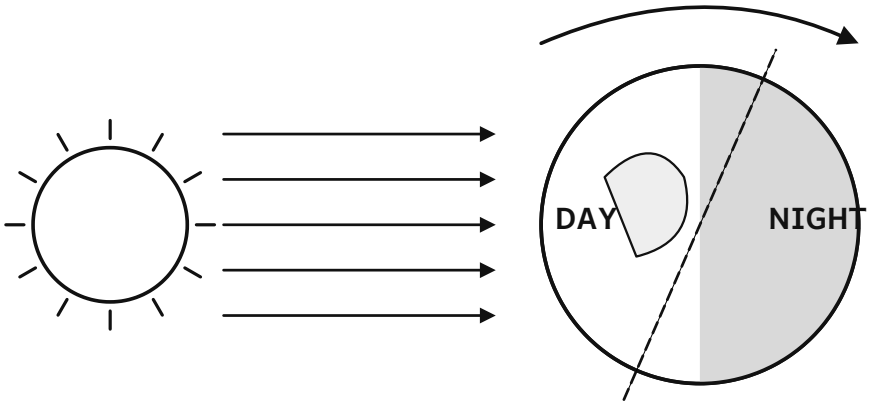
EARTH AND SPACE

Day, Night, Seasons and the Moon

The spinning Earth, the Sun's path, seasons, and the phases of the Moon



The Earth spins on its axis once every 24 hours.

In this pack

- **Learn it:** Day, night, seasons and the Moon ★☆☆
- **Fact check:** Earth, Sun and Moon ★☆☆
- **Think it through:** Explaining patterns ★★☆
- **Investigate:** Make a shadow stick ★★☆
- **Real-world case study:** The Moon and the fishers ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Day, night, seasons and the Moon

Name: Class: Date:

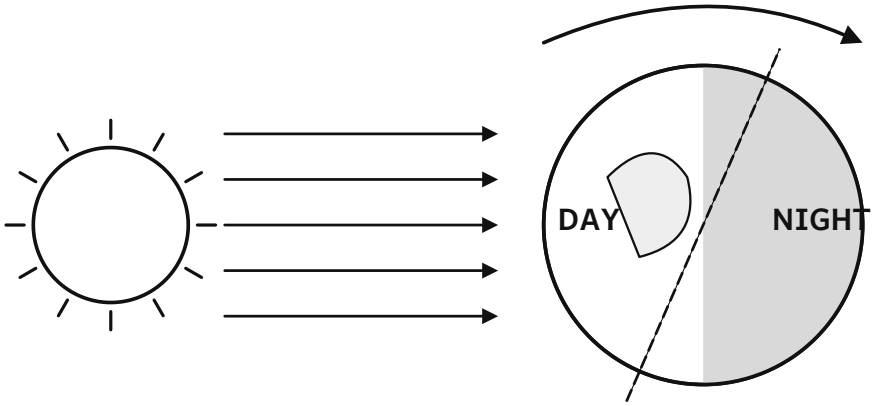
Learning goals

- Explain day and night using the spinning Earth.
- Explain why we have seasons.
- Describe the phases of the Moon and why they matter.

Every morning the Sun seems to rise in the east, and every evening it seems to set in the west. But the Sun is not really moving across the sky. **The Earth is spinning!**

Day and night

The Earth **spins** (rotates) on its **axis**, an imaginary line through the North and South Poles. It turns once every **24 hours**. The side facing the Sun has **day**; the side facing away has **night**.

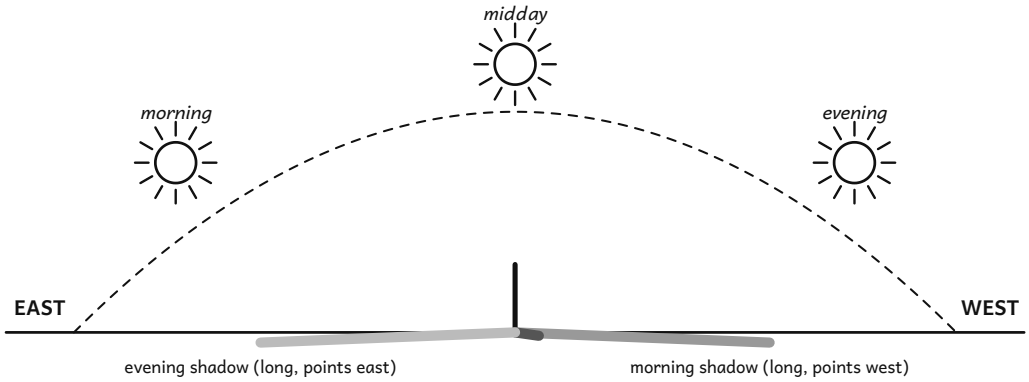


The Earth spins on its axis once every 24 hours.

Day and night.

The Sun across the sky

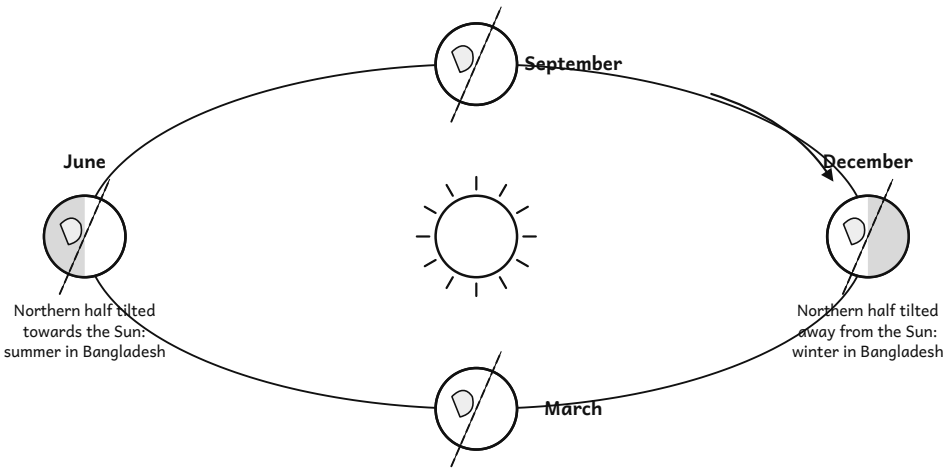
Because the Earth spins, the Sun seems to move from **east** to **west**. A **shadow stick** or **sundial** shows this: the shadow is long in the morning, short at midday and long again in the evening.



A shadow stick through the day.

A year and the seasons

The Earth also **travels around** (orbits) the Sun. One trip takes a **year**: 365¼ days. The Earth is **tilted**. When our part of the Earth leans towards the Sun, the Sun is **high** in the sky, its light is strong and we have **summer**. When we lean away, the Sun is **lower** and we have **winter**.

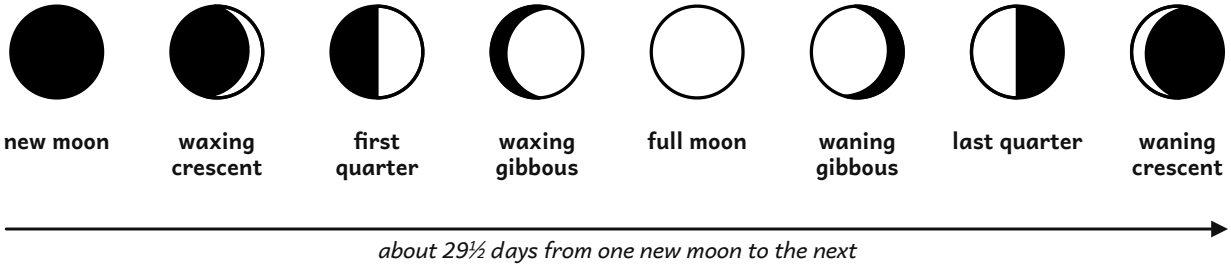


The tilted Earth travels around the Sun.

Seasons affect living things: farmers plant crops at the right season, some birds fly to Bangladesh in winter, and trees grow new leaves in spring.

The Moon and its phases

The Moon travels around the Earth about once a month. It has no light of its own; it **reflects** sunlight. As it moves, we see different amounts of its lit side. These shapes are called **phases**.



The phases of the Moon.

The Moon matters: its pull makes the **tides** go up and down, which fishers watch carefully. The Islamic months, and the dates of **Eid**, begin when the new crescent moon is seen. Some sea animals, like corals, lay eggs at the full moon.

Bangladesh connection

Bangladesh has **six seasons** (*ritu*): grishma (summer), barsha (rainy), sharat (autumn), hemanta (late autumn), sheet (winter) and basanta (spring).

Key words

rotate	spin around
axis	an imaginary line the Earth spins around
orbit	travel around
season	part of the year with its own weather
phase	the shape of the lit part of the Moon we see
tide	the sea going up and down each day

FACT CHECK · B2 · DAY, NIGHT AND MOON

Earth, Sun and Moon

★☆☆
Grade 3

Name: _____ **Class:** _____ **Date:** _____

Learning goals

- Recall key facts.
- Name Moon phases.

1 Fill in the gaps. [4]

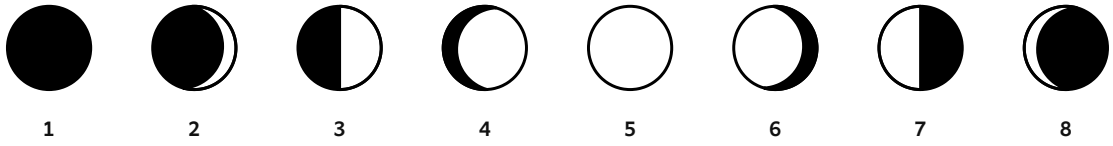
WORD BANK

Sun 24 year week Earth night

The Earth spins once every _____ hours. This makes day and _____.

The Earth goes around the Sun once a _____. The Moon goes around the _____.

2 Name the Moon phases. [4]



about 29½ days from one new moon to the next

1	2	3	4	5	6	7	8

3 True or false? [3]

Statement	True	False	If false, correct it
1. The Sun moves around the Earth every day.			
2. The Moon makes its own light.			
3. Shadows are shortest at midday.			

THINK IT THROUGH · B2 · DAY, NIGHT AND MOON

Explaining patterns

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Explain day and night and seasons.

1 **When it is day in Bangladesh, it is night in America. Why?** [2]

2 **Why is it hotter in summer than in winter?** [2]

3 **Draw a shadow stick at 9 am, 12 noon and 4 pm.** [3]

Shadow stick

INVESTIGATE · B2 · DAY, NIGHT AND MOON

Make a shadow stick

★★☆
Grade 3

Name: Class: Date:

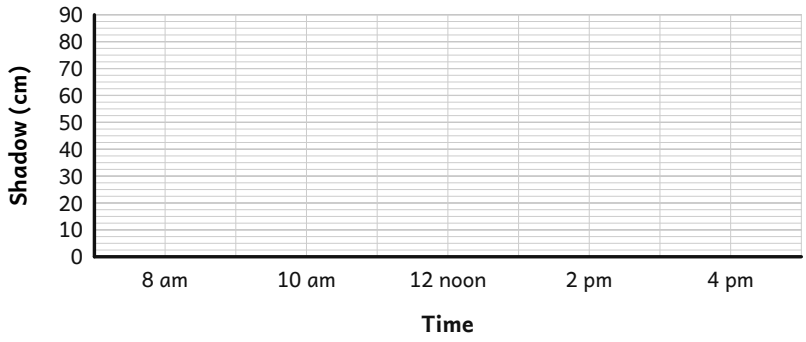
Learning goals

- Measure shadows through the day.
- Find a pattern.

Push a stick into the ground in a sunny place. Measure the length of its shadow every two hours.

1 **Here are Class 3A's results. Draw a bar chart.** [3]

Time	8 am	10 am	12 noon	2 pm	4 pm
Shadow (cm)	80	40	15	35	75



2 **When was the shadow shortest? Why?** [2]

.....

.....

3 **Predict the shadow at 5 pm.** [1]

.....

REAL-WORLD CASE STUDY · B2 · DAY, NIGHT AND MOON

The Moon and the fishers

★★★ Grade 3

Name: **Class:** **Date:**

Learning goals

- Learn why the Moon matters to people.

Science | Work | Culture

Fishers on the coast of Bangladesh watch the Moon carefully. At **full moon** and **new moon**, the tides are very high and very low. These are called **spring tides**. The fishers plan when to go out and when to bring their boats in.

Families also watch for the thin **crescent moon** after the new moon. When it is seen, a new month of the Islamic calendar begins, and everyone knows when **Eid** will be celebrated.

Based on tides and moon sighting in Bangladesh.

1 **Why do fishers watch the Moon?** [2]

2 **Which Moon phase begins a new month in the Islamic calendar?** [1]

3 **Draw the Moon tonight, or the last time you saw it. Which phase was it?** [2]

The Moon I saw

REVIEW AND CHALLENGE · B2 · DAY, NIGHT AND MOON

Show what you know

★★★
Grade 3

Name: Class: Date:

Learning goals

- Use Earth and Moon words.

1 Find the words.

J	S	F	E	G	R	V	G	U
T	N	P	N	W	A	D	A	O
K	C	I	I	W	S	A	F	D
F	Y	K	G	N	I	Y	L	W
G	R	D	I	H	D	M	R	M
T	I	D	E	S	T	K	U	O
E	C	S	E	A	S	O	N	O
O	R	B	I	T	L	U	D	N
U	K	R	T	H	F	C	K	O

day moon night orbit season spin tide

2 Why do we have day and night? [1]

◯ the Sun goes around the Earth ◯ the Earth spins

◯ clouds cover the Sun

3 How long does the Earth take to go around the Sun? [1]

◯ one day ◯ one month

◯ one year

4 Challenge. Use a torch and a ball to explain day and night to a friend. Draw how you would set it up. [2]

My model

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain day and night	○	○	○
make a shadow stick	○	○	○
explain seasons	○	○	○
name the Moon phases	○	○	○
say why the Moon matters	○	○	○